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16 October 1967
SUPERSEDING
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21 March 1967
(See 6.1)

MILITARY SPECIFICATION

TEST PROCEDURES FOR MASK PROTECTIVE ND MARK V

1. SCOPE

1.1 This specification covers test procedures for ND MARK protective masks for use with MIL-M-17522.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect, on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATION

MILITARY
MIL-M-17522 - Mask, Protective ND Mark V.

DRAWINGS

MILITARY

D-1976	- Rough Handling Device.
D-1979	- Light Transmission Device for Protective Mask Lens.
D-1982	- Pull Device for ND Mk V Canister Holders.
F846	- NRL-E3 Smoke Penetration Meter, Indicator Unit, Layout.
F854	- Smoke Penetration Meter E-3.
A880	- Smoke Cell and Phototube Holder, Model 3.
AA-850	- NRL-E3 Smoke Penetration Meter, Indicator Unit, Circuit Diagram.
E-136-42-750-1	- Q14 Gas Mask Leakage Tester.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Measurement of the ND Mk V canister holder to facepiece bond.

3.1.1 Outline of method. The procedure shall consist of the following:

(a) Determining the strength of the canister holder to facepiece of the protective mask.
(b) As a result of the test, accepting or rejecting the facepiece.

3.1.2 Apparatus. The apparatus shall be as follows:

(a) Scott tester with a capacity of at least 100 pounds.
(b) Canister holder clamps as shown on D-1982.

3.1.2.1 Preparation of apparatus. Preparation of the apparatus shall be as follows:

(a) Adjust the speed of the Scott tester to the specified value.
(b) Replace the standard clamps with the canister holder clamps shown on D-1982.
(c) Adjust the balance weights so that the tester will read zero at no load.

3.1.3 Operation. The operation shall be as follows:

(a) Secure the canister holders in the clamps and orient the facepiece as shown on D-1982. Facepieces from which samples for the lens bond strength test have been cut are adequate for this purpose.
(b) Start the tester and allow it to run to its limit.

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(c) If neither canister holder pulls loose at less than the number of pounds pull specified, accept the facepiece.
(d) If either canister holder pulls loose at less than the number of pounds pull specified, reject the facepiece.

3.2 Leak-testing the ND Mk V facepiece.

3.2.1 Outline of method. The procedure shall consist of the following:

(a) Determine the leakage of the facepiece.
(b) On the basis of this determination and MIL-M-17522 accept or reject the facepiece.

3.2.2 Apparatus. The apparatus shall consist of the following:

(a) One Q14 gas mask leakage tester assembly as shown on E-136-42-750-1.
(b) One vacuum pump with a capacity of 16 liters of air per minute at a pressure differential of 2 inches of water.
(c) One flowmeter with a capacity of 16 liters of air per minute.
(d) One manometer with a capacity of one inch of water.
(e) One air pressure regulator capable of controlling the pressure to plus or minus 0.1 pound (a Moore Nullmatic Pressure Regulator Model 40-15, or equal).

3.2.3 Reagents and materials. Reagents and materials shall be as follows:

(a) Technical grade dioctylphthalate (DOP).
(b) A filtered air supply.
(c) A source of clean dry air at 4 plus or minus 0.1 pound pressure.

3.2.4 Preparation of apparatus. Preparation of apparatus shall be as follows:

(a) Fill the generator with DOP. The level of the liquid should be 1/8 plus or minus 1/8 inch above the top of the collar on the Laskin nozzle.
(b) Turn on the controlled air supply specified in 3.2.3. (c) and allow the generator to run two minutes before using. If, at any time, the generator is allowed to sit idle for more than five minutes it should be flushed out again before using.
(c) Prepare the light scattering cell and indicator unit as specified in 3.8.
(d) Connect the flowmeter into the vacuum side of the dummy head between the light cell and the vacuum source.
(e) Insert a valve (or a pinchclamp on the rubber hose) between the filtered air supply and the facepiece connection.
(f) Connect the manometer to the tube leading through the dummy head to inside of the facepiece.
(g) Check all connections for leaktightness.

3.2.5 Operation. The operation shall be as follows:

(a) Mount the facepiece on the dummy head in such a manner to assure an airtight seal.
(b) Wet the outlet valve to ensure an airtight seal.
(c) Connect the filtered air supply to the facepiece.
(d) Start the vacuum pump.
(e) Adjust the vacuum pump and the valve specified in 3.2.4(e) until the airflow through the facepiece is 16 parts per million (PPM) at 1 inch of water pressure differential.
(f) Direct a jet of DOP smoke from the aerosol generator.
    (1) Around the lens to faceblank bond.
    (2) Around the speech-diaphragm outlet-valve assembly to faceblank bond.
    (3) Around each of the canister holders to faceblank bonds.
    (4) Around the speech-diaphragm seal. This may be done by directing a jet of smoke through several of the holes in the speech-diaphragm-outlet-valve holder.
(g) During the operation specified in 3.2.5. (f) operate the light scattering cell and indicator unit as specified in 3.8 to determine the quantity of smoke which leaks through the various bonds.
(h) Before rejecting a mask for leakage the operator shall assure himself that the leak is due to one of the bonds under test and is not due to any one of the following:
    (1) Failure to seal the mask against the headform.
    (2) Failure to seal the outlet valve.
    (3) Failure to seal all connections.
(i) If the facepiece meets the minimum requirements for leakage, accept the facepiece.
(j) If the facepiece does not meet the minimum requirements for leakage, reject the facepiece.

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3.2.6 Shutdown. The procedure shall be as follows:

(a) Turn off the air supply to the aerosol generator.
(b) Shut down the light scattering cell and the indicator unit as specified in 3.8.

3.2.7 Maintenance. The maintenance shall be as follows:

(a) The level of DOP in the aerosol generator shall not be allowed to fall below the lower edge of the collar on the Laskin nozzle.
(b) The air supply for the aerosol generator shall be checked occasionally to make certain that it is clean and dry.
(c) The calibration of the flowmeter shall be checked occasionally.
(d) The maintenance of the light scattering cell and the indicator unit shall be as specified in 3.8.

3.3 Rough handling the ND-C-2 canister.

3.3.1 Apparatus. The apparatus shall be a rough handling device as shown on D-1976.

3.3.2 Definitions. Definitions shall be as follows:

(a) Horizontal arm. The horizontal arm is designed to rough handle a canister in a horizontal position. It shall be as shown on items 5 and 32, sheet 1 of D-1976.
(b) Vertical arm. The vertical arm is designed to rough handle a canister in a vertical position. It shall be as shown on items 12 and 33, sheet 1 of D-1976.
(c) Canister holder. The canister holder is designed to hold the ND-C-2 canister during the rough-handling procedure. It shall be as shown on item 1, sheet 2 of D-1976.
(d) Lock ring. The lock ring is designed to secure the ND-C-2 canister in the holder during the rough handling procedure. It shall be as shown on item 2, sheet 2 of D-1976.
(e) Automatic stop. The automatic stop is designed to hold the arm(s) until the moment of release and to control accurately the distance through which the arm is permitted to fall. It shall be as shown on sub-assembly 15, sheets 1 and 2 of D-1976.

3.3.3 Operation. The operation procedure shall be as follows:

(a) Horizontal rough handling.
  (1) Place the canister in the horizontal arm of the rough handling device. Place the outlet or "apron side" of the canister down, that is, in contact with the inside flange of the canister holder. Secure the lock ring against the inlet side of the canister.
  (2) Raise the horizontal arm past the automatic stop.
  (3) Return the horizontal arm to the automatic stop and allow it to rest thereon.
  (4) Release the automatic stop.
  (5) Repeat steps 2, 3 and 4 until the canister has been rough handled the specified number of times.
(b) Vertical rough handling.
  (1) Place the canister in the vertical arm of the rough handling device. Place the outlet or "apron side" of the canister towards the center of the rough handling device, that is, in contact with the inside flange of the canister holder. Secure the lock ring against the inlet side of the canister.
  (2) Rise the vertical arm past the automatic stop.
  (3) Return the vertical arm to the automatic stop and allow it to rest thereon.
  (4) Release the automatic stop.
  (5) Repeat stops 2, 3 and 4 until the canister has been rough handled the specified number of times.
  (6) Mark the canister before its removal from the rough handling device to show orientation.

3.3.4 Maintenance. The rough handling device should be checked occasionally for loose parts and for signs of excessive wear; moving parts shall be greased as necessary.

3.4 Determination of physical distortion in the lens.

3.4.1 Outline of method. The procedure shall be as follows:

(a) Determine whether or not the lens installed in the facepiece meets the minimum requirements for physical distortion.
(b) On the basis of this determination, accept or reject the facepiece.

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3.4.2 Apparatus. The apparatus shall be as follows:

(a) A straightedge 2-3/4 inches long.
(b) A straightedge 3-1/8 inches long.

3.4.3 Preparation of sample. Preparation of samples shall be as follows:

(a) With the lens installed in the facepiece locate the following lines. (Do not mark the lens. The position should be indicated by marking the rubber faceblank adjacent to the lens.)
(1) The vertical centerline of the lens.
(2) A line parallel to, and two inches to the right of the center line.
(3) A line parallel to, and two inches to the left of the center line.

3.4.4 Operation. The procedure shall be as follows:

(a) Lay the 2-3/4 inch straightedge on the vertical center line.
(b) Determine the maximum distance between surface of the lens along the vertical centerline and the straightedge.
(c) Place the 3-1/8 inch straightedge successively along each of the lines parallel to, and two inches away from the center line.
(d) Determine the maximum distance between the surface of the lens along either of these lines and the straightedge.
(e) If the maximum distance between the surface of the lens and the straightedges does not exceed the maximum deviation permitted by MIL-M-17522, accept the facepiece.
(f) If the maximum distance between the surface of the lens and the straightedges exceeds the maximum deviation permitted by MIL-M-17522, reject the facepiece.

3.5 Pull test for (inspection apparatus Q22 tester) C-2 canister apron staking strength.

3.5.1 Purpose. The pull test device for the ND-C-2 canister apron tests the staking strength between the rubber apron and the canister. The device determines whether the apron can withstand a force of 25 pounds' pull.

3.5.2 Installation.

3.5.2.1 The tester shall be placed in a position where there is sufficient space for operation and maintenance.

3.5.2.2 An air pressure source of approximately 90 psi and a power source of 110 volts AC shall be available.

3.5.2.3 The oiler, item 1 shown on figure 1, on the back of the machine shall be filled 3/4 full with a good SAE 20 lubricating oil. This is done by removing filler cap, item 2 shown on figure 1, making sure the pressure is off, and filling to a 3/4 level.

3.5.2.4 The oiler needle valve, item shown on figure 1, shall then be opened about 2 turns.

3.5.2.5 Connecting air line. The air control valve, item 5 shown on figure 2, shall be closed and opened only after pressure has been applied to the line. Check to make sure pressure regulator valve, item 15 shown on figures 1 and 2, is closed. This valve shall always be closed when connecting or disconnecting the air line to avoid damage to the pressure gage, item 14 shown on figures 3 and 4. The air line shall be connected to tester by means of air inlet fixture, item 4 shown on figures 2 and 3 on right hand side. Open air control valve. Open pressure regulator valve, until pressure gage reads 33 psi. Connect electric power cord to electric source.

3.5.3 Press release button, item 6 shown of figure 4. This will bring piston rod, item 7 shown on figure 3, to foremost position.

3.5.3.1 The control rack, item 8 shown on figure 3, shall be checked to see that it is also as far forward as possible. If end of adjustment screw, item 9 shown on figure 3, on rear part of the control rack is not in contact with the piston support, item 10 shown on figure 3, push the control rack forward by hand, but do not change the setting of the adjustment screw. It has been factory set. This screw limits the forward movement of the control rack. During the normal course of operation and testing of the canister apron, the control rack locates itself automatically.

3.5.4 Operation. The procedure shall be as follows:

(a) Press canister with its apron upward between the two hard rubber rollers, item 11 shown on figure 3, on front of the tester. Push the canister in as far as it will go.

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(b) Check that the apron slides between the clamping cam, item 12 shown on figure 3, and the lower jaw of the pull mechanism, item 13 shown on figure 3.
(c) Check that the control rack is as far forward as it will go.
(d) Check pressure gage to see if it reads 33 psi. If not, remove the canister and adjust the pressure regulator valve, item 15 shown on figure 1, at rear of tester, pressing the release button after each adjustment to release any built up pressure.
(e) Press operate button, item 16 shown on figure 4.
(f) Leave apron in stretched position until pressure gage reaches 33 psi, then press release button. If anything other than the force exerted by the canister apron causes the piston to stop moving before the gage reaches 33 psi. (for example, the jaw of the pull mechanism, item 13 shown on figure 3, strikes and is stopped by the piston support, item 10 shown on figure 3, the test shall be void.
(g) If the force of the piston on the apron pulls the apron from the canister, reject the canister.
(h) If the force of the piston fails to pull out the apron, turn canister so that another part of the apron is in the pull mechanism and repeat the test specified in 3.5.4, item (e) through (h).
(i) If the canister apron passes the test at several points around the canister, it is acceptable.
(j) Record results.

3.5.5 Maintenance. The maintenance shall be as follows:

(a) If it is ever necessary to disconnect tester from air supply, shut needle valve off, item 5 shown on figure 2, and disconnect it at joint, item 18 shown on figure 2, nearest the tester.
(b) Check the oil level in oiler at least once a week. Keep it at least 3/4 full.
(c) On the right side of the tester is a hole through which can be seen the air separator, item 19 shown on figure 2. Check periodically to see if there is water in the separator. If there is, open petcock, item 17 shown on figure 2 at bottom of separator. Air pressure will blow water out. The petcock can be reached through opening under side panel of the tester.

3.6 Measurement of coloration and physical distortion.

3.6.1 Outline of method. The procedures shall consist of the following:

(a) Lens coloration.
  (1) Determining the light transmission of the facepiece lens.
  (2) Heating the facepiece under controlled conditions.
  (3) Redetermining the light transmission of the facepiece lens.
  (4) As a result of the test, accepting or rejecting the facepiece.
(b) Physical distortion of lens.
  (1) Heating a facepiece containing a nondistorted lens under controlled conditions.
  (2) Determining if physical distortion is present in the lens.
  (3) Accepting or rejecting the facepiece as a result of the test.

3.6.2 Apparatus. The apparatus shall consist of the following:

(a) Oven. Constant temperature, capable of operating at 75°C. plus or minus 1°C.
(b) Facepiece containers. These are made from rectangular, tinned metal, solvent containers with dimensions 6-5/8 inches wide by 4-1/8 inches deep by 9-1/2 inches high (nominally one-gallon size). Remove "pour spout" end of the solvent container to provide an opening which is 6-5/8 by 4-1/8 inches. Retain the rolled edge when making the opening. Before use, heat the containers at 100°C. for 24 hours.
(c) Spectrophotometer. Beckman, Model DU, or equal, equipped with special adaptor for determining the lighttransmission of the gas mask lens. D-1979 presents a diagram of the adaptor. The apparatus consists mainly of a two-sectioned tube which provides a light path. At the junction of the two tubes are two flanges, one moveable, the other stationary. The other parts of the apparatus are mounting plates, for securing the adaptor to the spectrophotometer, and various supporting members. The design of the adaptor permits lighttransmission measurements to be made on an intact lens mounted in a facepiece.

3.6.3 Reagents and materials.

(a) Foil. Aluminum, of about 0.001 inch thickness. One rectangular piece, 9 by 7 inches, shall be required for each facepiece container.
(b) Cloth. About one square yard of black, opaque cloth is placed over the sample and adaptor to exclude stray light during transmission measurements.

3.6.4 Preparation of apparatus, lens coloration test. After the special adaptor for the spectrophotometer has been assembled, it shall be installed in the following manner:

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(a) Detach phototube housing from spectrophotometer by removing assembly screws.
(b) Remove cell holder and cell compartment and cover.
(c) With machine screws, attach the smaller of the two end plates of the adaptor to the cell compartment mounting block of the spectrophotometer. Be sure that the guide pins of mounting block engage the corresponding holes in the end plate of the adaptor.
(d) Engage guide pins of phototube housing with corresponding holes in the other end plate and attach phototube housing to end plate with assembly screws.
(e) Place a block of suitable size under the phototube housing to provide a rigid support.
(f) For light transmission measurements, adjust the spectrophotometer as follows:
(1) Wavelength scale set at 450 millimicrons (mu).
(2) Sensitivity adjustment set three turns from clockwise limit.
(3) Tungsten filament lamp light source in use.
(4) Ultraviolet sensitive phototube in use.
(5) Instrument adjusted to 100 percent lighttransmission against air. Vary slit width to obtain this adjustment.

3.6.5 Procedure, lens coloration test.

(a) Remove head harness and canisters from mask, invert facepiece, and, with the outside mask surface adjacent to photocell housing, insert facepiece between the two flanges of the adaptor until the desired lens area is positioned in the light path. Slide the moveable flange against the lens so that the lens is firmly held. Drape opaque cloth over sample to exclude stray light.
(b) Determine lighttransmission of the sample with the spectrophotometer adjust as outlined in 3.6.4(f). Remove facepiece from adaptor. If the value obtained meets the requirements of MIL-M-17522 for the initial light-transmission value of the lens, continue the procedure beginning with item (c).
(c) Insert facepiece, having a nondistorted lens, "chin first" into facepiece container so that the speech diaphragm assembly faces the broadest wall. The facepiece will fit snugly but should retain its normal shape when in place. If the top of the speech diaphragm assembly presses against the container wall because of insufficient clearance within the container, the wall should be bent outward slightly to release this pressure point. Caution. Make certain that the lens is not in contact with metal. Avoid any contact of the lens with rubber except, of course, where the lens is bonded to the facepiece.
(d) Place a piece of 9 by 7 inch aluminum foil over the open end of the facepiece container so that about one inch overlaps the container walls. Fold the overlapping foil down over the outside of the container to form a snug fitting cover. The foil cover shall be free of holes during the subsequent heating period.
(e) Place container, with foil cover up, in constant temperature oven.
(f) Heat container at 75°C. for 24 hours.
(g) Remove container from oven and allow mask to cool.
(h) Determine physical distortion of the lens. If the condition of the lens conforms to MIL-M-17522, continue with item (i).
(i) Determine the lighttransmission value of the lens as specified in items (a) and (b).

Accept facepiece if the lighttransmission value is in conformance with MIL-M-17522.

3.6.6 Maintenance. Maintenance shall be as follows:

(a) The temperature of the oven(s) shall be checked, and adjusted if necessary, prior to heating the masks.
(b) The interior surfaces of the sample containers in which the masks are heated may become contaminated with prolonged use. To minimize the possibility of error from this source, it is advisable to heat the empty containers, occasionally, at 100°C., for 24 hours.

3.7 Measurement of the lens bond strength.

3.7.1 Outline of method. The procedure shall be as follows:

(a) Cut test specimens from the facepiece being evaluated.
(b) Determine the strength of the lens bond using a Scott testing machine.
(c) On the basis of this determination, accept or reject the facepiece.

3.7.2 Apparatus. The apparatus shall be a Scott testing machine, Model J, or equal..

3.7.3 Preparation of sample.

(a) Cut facepiece along line AA of figure 5.
(1) Use upper portion of facepiece for lens bond strength test.
(2) Use lower portion of facepiece for canister holder pull test.

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(b) Cut a minimum of four test specimens from the facepiece (see figure 5).
(1) Each specimen shall be 1 plus or minus 1/32 inches wide.
(2) Each specimen shall be at least two inches long, approximately half lens and half rubber.
(3) Each specimen shall be cut so that the centerline of the specimen is perpendicular to the tangent to the bond at that point.

3.7.4 Preparation of apparatus. Preparation of apparatus shall be as follows:

(a) Set the Scott tester at the specified rate of travel.
(b) Equip the tester with one fabric clamp and one rubber clamp.
(c) Adjust the balance weights so that the tester reads zero at no load.

3.7.5 Operation. The operation shall be as follows:

(a) Clamp a test specimen into the tester. Make certain that the rubber clamp does not clamp on the lens-bond area.
(b) Start the tester and allow it to run until the test specimen is pulled apart.
(c) Record the force (pounds) required to separate the specimen.
(d) Repeat (a), (b) and (c) for each test specimen from the facepiece.
(e) Average the results.
(f) If the lens-bond strength meets the requirements of the specification, accept the facepiece.
(g) If the lens-bond strength does not meet the requirements of the specification, reject the facepiece.

3.8 Operation of the NRL light scattering meter.

3.8.1 Apparatus. The apparatus shall be as follows:

(a) One indicator unit as shown on F846 and AA850.
(b) One smoke cell and phototube holder as shown on F854 and A880.
(c) One 6 volt, 50 watt, constant voltage transformer.

3.8.2 Definitions. Definitions shall be as follows:

(a) Standard light leak. A standard light leak is one which has been calibrated. This leak is utilized in standardization of the instrument under field conditions. The light leak is essential in obtaining reproducible results over extended time intervals.
(b) Optical filter. An optical filter is a perforated metal disc used to reduce by a known amount the intensity of light reaching the phototube.
(c) Scale switch. The scale switch is used to select the proper scale range. The scale multipliers are selected to give scale ranges of 0-100 percent (optical filter in light beam) 0 to 1 percent and 0 to 0.1 percent. For the two latter scales it is necessary to operate with the optical filter out of the light beam. If the cell contains a standard light leak, the shield shall be inserted on the two lower scales.
(d) Sensitivity controls. The sensitivity control is used to adjust the microammeter reading without affecting the output voltage or the stray light compensation.
(e) Scale adjustment control. The scale adjustment control is used to adjust the zero reading of the microammeter when the scale switch is set on scale zero.
(f) Stray light compensation. The stray light compensator controls a back EMF which opposes the dark current and the stray light current from the phototube. It is used to adjust the zero reading of the microammeter when the smoke cell is filled with clean air and the scale switch is set on 0.1 percent.
(g) Output voltage. The output voltage regulator is used to adjust the 100 percent reading of the microammeter when the smoke cell is filled with an aerosol, the optical filter is in the light beam and the scale switch is set on 100 percent. If a standard light leak is used, the output voltage is used to adjust the microammeter reading to the predetermined value of the standard light leak.

3.8.3 Preparation and calibration of a standard light leak.

(a) A suitable light leak shall be prepared by cutting a hole approximately 1/16 inch in diameter in the center of the black spot on the condenser lens. The hole shall then be covered by translucent paper to diffuse the light. When light scattering measurements are being made, the hole shall be covered by a special shield inserted in place of the optical filter.
(b) Calibration of the light leak is accomplished by using a dioctylphthalate aerosol of 0.3 micron droplet diameter and a mass concentration of 80 to 100 micrograms per liter. This aerosol is passed through the cell and the output voltage adjusted to a 100 percent setting on the meter. Filtered air shall then be introduced to the cell and the optical

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filter removed. In place of the optical filter the light shield is inserted and the scale selector switch turned to the 0.1 percent scale. The meter reading shall be then adjusted to zero using the stray light compensation. The light shield shall then be removed and the meter reading and selector switch position recorded as the standard light leak.

3.8.4 Preparation of apparatus (using a standard light leak). The preparation shall be as follows:

(a) Turn the indicator unit on and allow it to warm up for approximately ten minutes.
(b) Turn the cell lamp on.
(c) Set the sensitivity control to the mid-position.
(d) Set the scale switch on scale zero.
(e) Turn the scale adjustment control until the microammeter reads zero.
(f) Flush the smoke cell with clean air.
(g) Set the scale switch to the appropriate scale for the standard light leak.
(h) Remove the shield from the standard light leak.
(i) As soon as the microammeter needle comes to rest, adjust the output voltage until the microammeter reads the predetermined value for the standard light leak.
(j) Replace the shield covering the standard light leak.
(k) Set the scale switch to 0.1 percent.
(l) Adjust the stray light compensation until the microammeter reads zero.
(m) Repeat steps (g) to (l) at least once and as many times as necessary to properly adjust the circuit.

3.8.5 Preparation of apparatus (without standard light leak). The preparation shall be as follows:

(a) Turn the indicator unit on and allow it to warm up for approximately ten minutes.
(b) Turn the cell lamp on.
(c) Set the sensitivity control to the mid-position.
(d) Set the scale switch on scale zero.
(e) Turn the scale adjustment control until the microammeter reads zero.
(f) Set the scale switch on 100 percent.
(g) Put the optical filter in the path of the light beam.
(h) Pass the aerosol through the smoke cell at the specific flow rate required by the test procedure. (Flow rates of 10 to 100 liter per minute are recommended.)
(i) As soon as the microammeter needle comes to rest, adjust the output voltage until the microammeter reads 100 percent.
(j) Flush the smoke cell with clean air.
(k) As soon as the microammeter reading is less than 5, remove the optical filter from the light beam.
(l) As soon as the microammeter reading has fallen below 20, turn the scale switch to 1 percent.
(m) As soon as the microammeter reading has fallen to approximately 10, turn the scale switch to 0.1 percent.
(n) As soon as the microammeter needle comes to rest, adjust the stray light compensation until the microammeter reads zero.
(o) Repeat steps (f) to (n) at least once and as many times as necessary to properly adjust the circuit.

3.8.6 Operation. The operation shall be as follows:

(a) Place the optical filter in the path of the light beam.
(b) Set the scale switch on 100 percent.
(c) Introduce the unknown aerosol into the smoke cell (see 3.8.5 (h)).
(d) If the microammeter reads less than 5, remove the optical filter.
(e) If the microammeter now reads less than 20, turn the scale switch to 1 percent.
(f) If the micorammeter now reads less than 10, turn the scale switch to 0.1 percent.

3.8.7 Shutdown. The shutdown shall be as follows:

(a) Flush the smoke cell with clean air.
(b) Turn the indicator unit and the cell lamp off.
(c) Set the scale switch on the scale zero.

3.8.8. Maintenance. The maintenance shall be as follows:

(a) Measurement of stray light and dark current.
(1) Stray light is defined as the percentage of unfiltered smoke that is indicated by the stray light reaching the phototube when filtered air is in the smoke cell. Dark current is defined as the current produced by all extraneous sources not accounted for by either stray light or scattered smoke. When making routine measurements of

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penetration, a daily check on the amount of stray light and dark current is recommended. To do this, set the amplification with unfiltered smoke in the cell to give full scale deflection with the scale switch set in the 100 percent position. Sweep the smoke cell with filtered air; pull out the neutral optical filter; and turn the stray light compensation knob as far as possible in the clockwise direction. Set the scale switch in the 0.1 percent position. The meter shall read the total leak, that is, the dark current plus the stray light. If the cell lamp switch is turned off, the reading will decrease by the number of units corresponding to the stray light, leaving a measure of the dark current. The stray light shall be less than 0.010 percent of the unfiltered smoke. If this is not the case, the lenses of the smoke cell shall be cleaned.

(b) Cleaning the lenses of the smoke cell.

(1) General. When the amount of stray light is greater than 0.010 percent of the unfiltered smoke, the lenses in the optical system are probably covered with a thin film of condensed smoke. The cleanliness of the back lens is more critical than that of the front lens. The indicator unit shall be turned off during the following procedures.

(2) Back lens. To clean the back lens, remove the six back screws and remove the back lens plate and phototube mount. The back lens will then be found on the bottom of the plate. After breathing upon the lens to moisten it, wipe away any condensed smoke film or other foreign material with a piece of lens paper. Replace the plate, the phototube mount, and the six mounting screws.

(3) Front lens. To find the front lens, remove the screws on the front flange of the smoke cell and remove the body of the cell. The front lens will then be exposed. After breathing upon the lens to moisten it, wipe away any condensed smoke film or other foreign material with a piece of lens paper, using care not to damage the black spot of paper on the lens. Replace the body of the smoke cell and the six screws. Remeasure the amount of stray light; if it is still greater than 0.010 percent of the unfiltered smoke, the optical line-up must be adjusted.

(c) Adjusting the optical line-up.

(1) To adjust the optical line-up for the minimum amount of stray light, first check the bulb alignment by means of the cell lamp adjusting screw at the base of the smoke cell. Next, loosen the three photo-mount adjusting screws on the smoke cell, suck filtered air through the smoke cell, and pull out the optical filter. With the aid of another operator, systematically move the phototube mount until a minimum of stray light is indicated by the meter. If these adjustments fail to reduce the stray light to less than 0.010 percent of the unfiltered smoke, the following alternative procedure shall be followed.

(2) By means of the six back lens screws remove the back lens plate. Ascertain that the back lens is clean. With the cell lamp on and the optical filter pulled out, place a small piece of white tissue paper against the hole in the diaphragm. If the spot of light is not centered in the hole, loosen the thumb screw at the base of the smoke cell and adjust the position of the cell lamp until the spot is centered. When these adjustments have been made, reassemble the smoke cell and adjust the position of the phototube mount minimum stray light by means of the phototube mount adjusting screws. Tighten the three phototube mount adjusting screws. Do not, however, make any further adjustment in the position of the cell lamp. Check that the adjustments just made have not changed the unfiltered smoke reading appreciably. Before making any further penetration measurements, set the zero by means of the stray light compensation knob.

(d) Replacing cell lamp. The 50 candlepower headlight bulb (Mazda number 1061, or equal) on the smoke cell shall be replaced when necessary. By loosening the cell lamp adjusting screw at the base of the smoke cell, the bulb position shall be adjusted so that a maximum reading for unfiltered smoke is obtained, and, at the same time, a minimum of stray light compensation is needed with filtered air going through the smoke cell and the optical filter pulled out.

(e) The indicator unit. Replace, if necessary, the 6V 1 Amp pilot light. If the indicator unit appears to be entirely dead, replace the 1 Ampere instrument fuse. If additional repairs are required, consult an electronic engineer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Not applicable.

5. PREPARATION FOR DELIVERY

5.1 Not applicable.

6. NOTES

6.1 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - SH
(Project 4240-N345)

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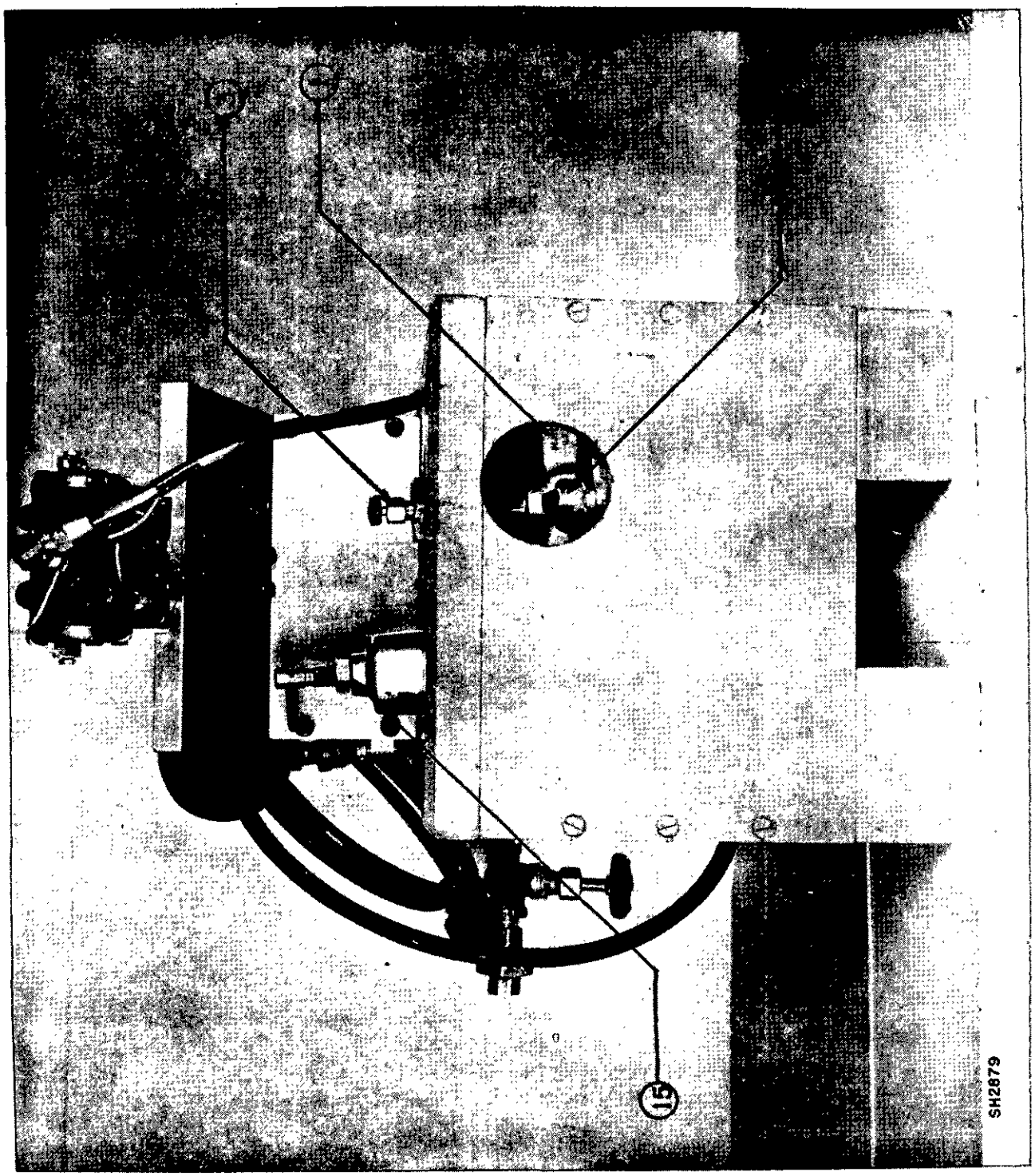


Fig. Pull test device.

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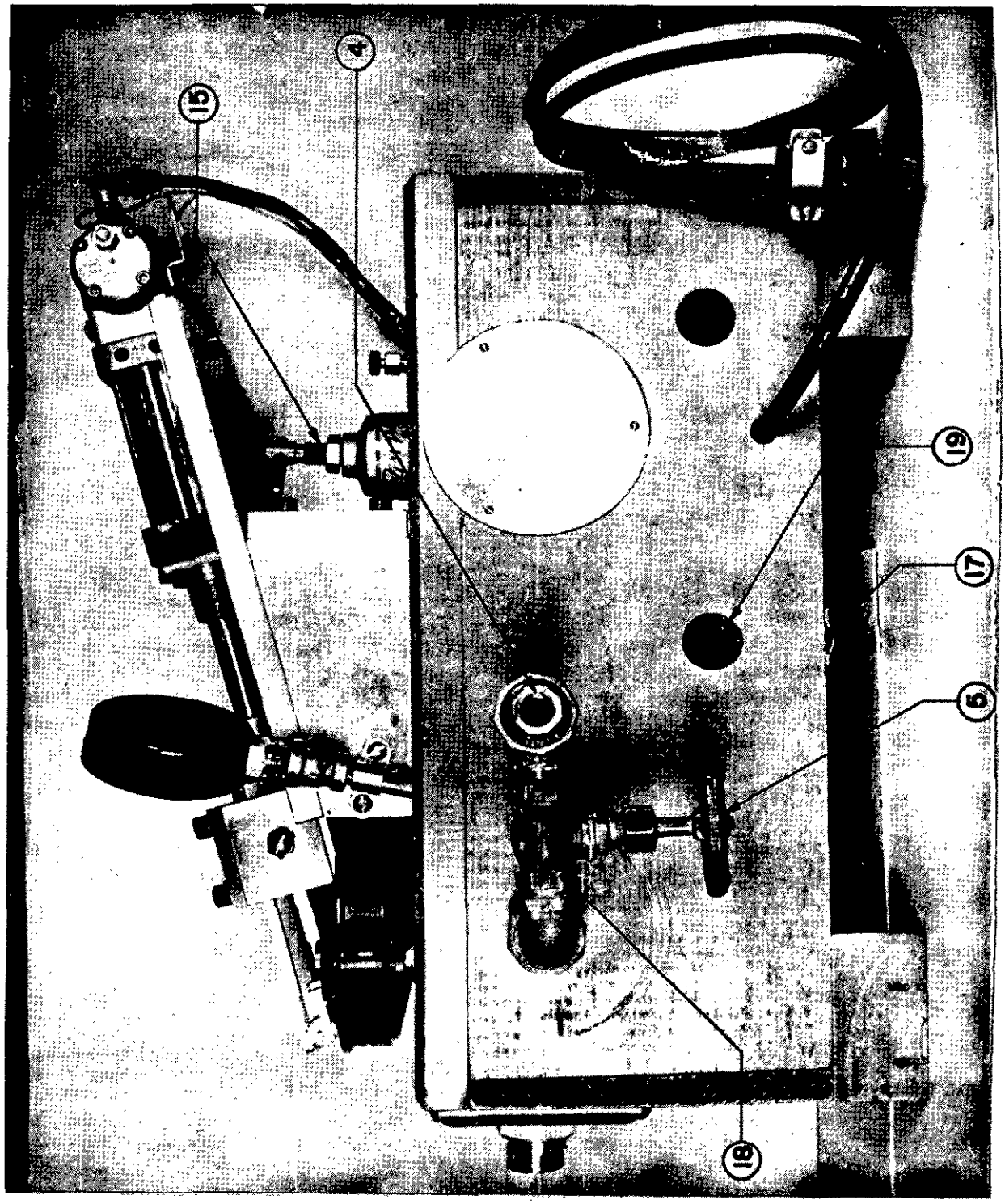


Figure 2 - Pull test device.

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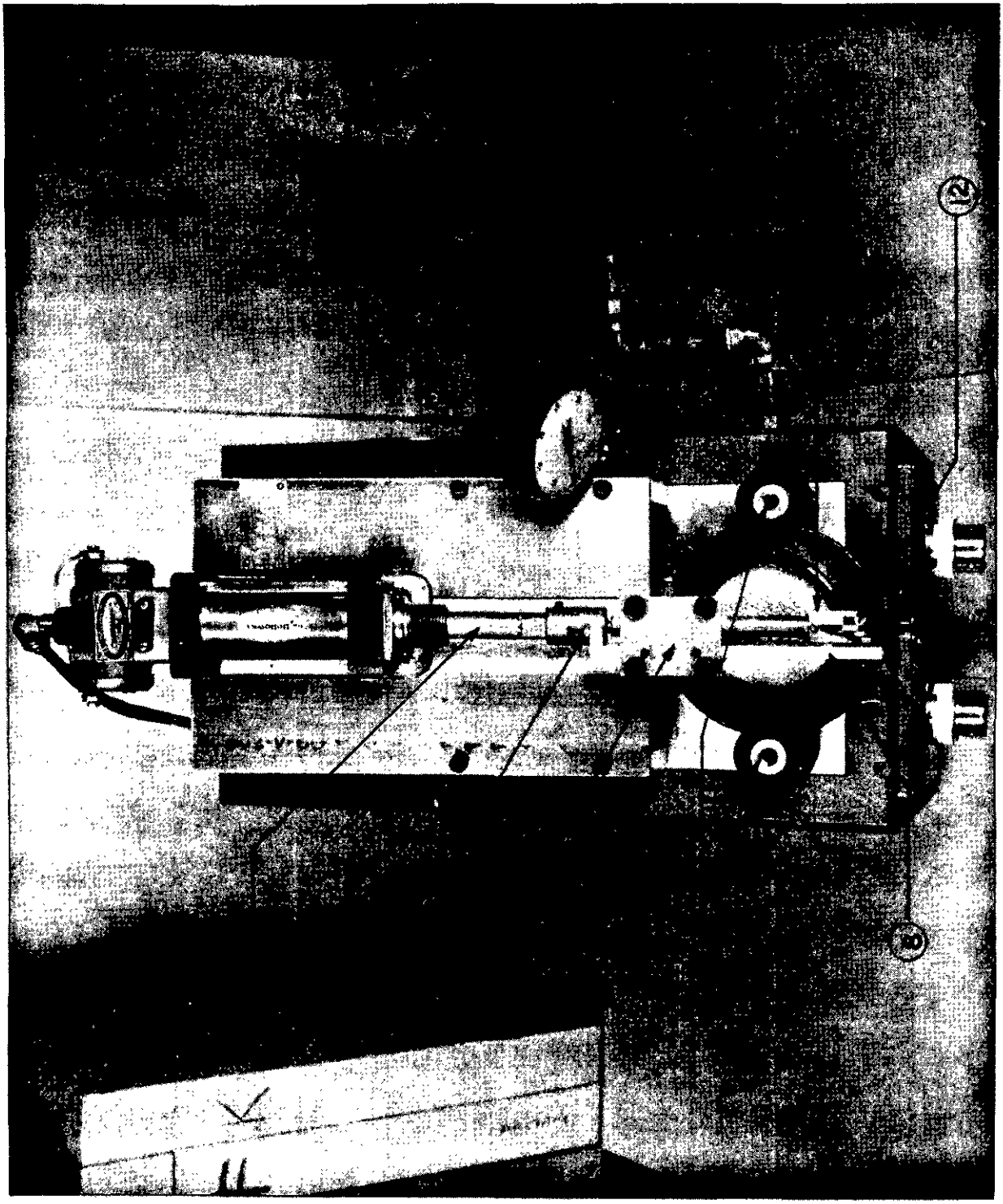


Figure 3 - Pull test device.

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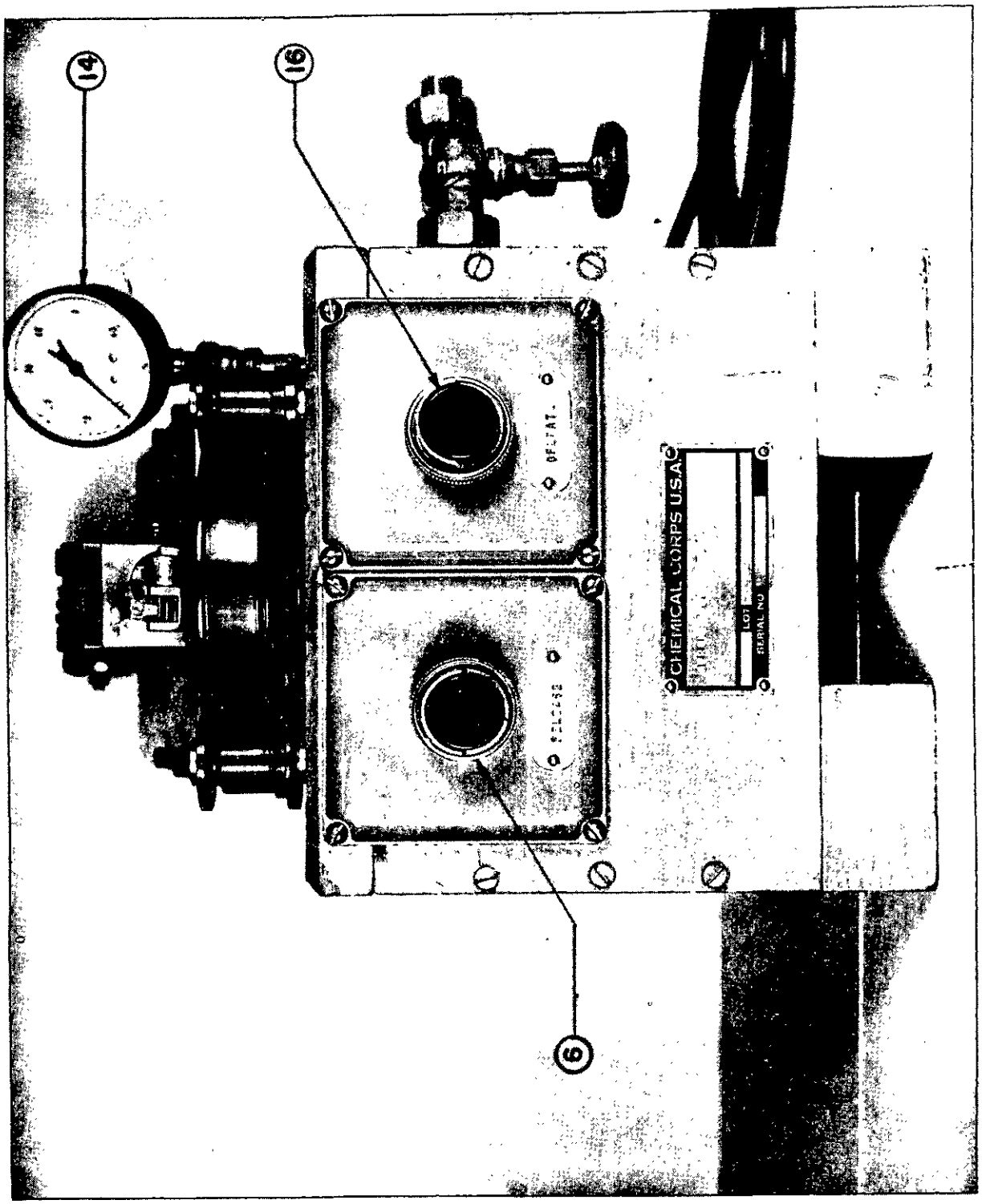


Figure 4 - Pull test device.

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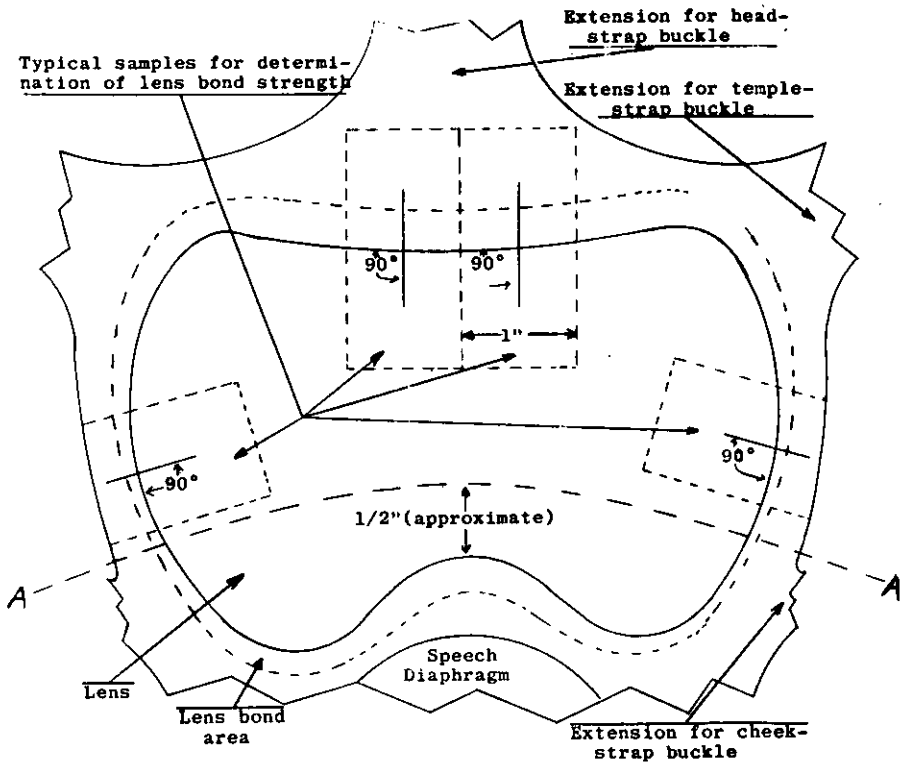


Figure 5 - ND-MK protective mask facepiece.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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